CARMA-Based Deseasonalization, Calibration and Pricing in Renewable Energy Markets

July 2, 2026

Abstract

1 Introduction

Electricity prices are increasingly exposed to weather and renewable generation. Temperature affects demand, while wind and solar production affect both the level and the intraday shape of spot prices. Energy contracts written on such markets therefore carry hybrid risk, since their value depends on the joint behavior of electricity prices and physical factors that are non-traded or only partially traded.

This paper develops an empirical CARMA framework for these joint risks. We first remove deterministic seasonal patterns from German day-ahead prices, temperature, wind capacity factors, and solar capacity factors. The resulting stochastic factors are then modeled with continuous-time autoregressive moving-average dynamics, which provide a state-space representation for calibration, dependence estimation, and pricing.

The preprocessing is factor-specific because the deterministic component differs across markets and weather variables. Electricity prices contain calendar and intraday effects, temperature has smooth daily and annual cycles, and wind capacity factors are naturally treated through a logit transformation. Solar requires an additional clear-sky normalization before the latent logit transformation, as night hours and moving sunrise and sunset times induce deterministic boundary effects.

The contribution is threefold. First, we build a consistent preprocessing pipeline for electricity, temperature, wind, and solar data. Second, we calibrate marginal CARMA models and recover their driving noise, allowing heavy-tailed Levy specifications when Gaussian innovations are insufficient. Third, we use the calibrated factors to estimate price-weather and price-renewable dependence channels and to price energy contracts exposed to these joint risks.

The paper is organized as follows. Section 2 describes the data transformations and the construction of the stochastic factors. Section 3 presents the marginal CARMA calibration and driver diagnostics. Section 4 estimates the dependence channels. Section 5 discusses the pricing implementation, and Section 6 concludes.

2 Data and Deseasonalization

2.1 Electricity Prices

Data. We use hourly German day-ahead electricity prices for the DE–LU bidding zone. The calibration sample runs from 1 January 2023 to 31 December 2025 and is indexed in UTC.

The 2021–2025 history is shown in Figure 1. The 2021–2022 period contains exceptional crisis spikes, while the 2023–2025 window keeps a recent regime with both positive and negative price spikes. Since the German spot market can contain negative prices, the price is first shifted before applying the logarithm:

$$\widetilde{S}_t = \log(S_t + \delta), \qquad \delta = 1000 \text{ EUR/MWh}. \tag{1}$$

The shift is only used to define a positive log-price factor. Physical prices are recovered by the inverse map $S_t = \exp(\widetilde{S}_t) - \delta$.

Deseasonalization. The price model is specified in the shifted log-price coordinate:

$$\widetilde{S}_t = \Lambda_t^P + X_t^P. \tag{2}$$

Here Λ_t^P is deterministic and X_t^P is the stochastic price factor later modeled by CARMA dynamics. We estimate Λ_t^P with a Paraschiv-style seasonal weight in the transformed coordinate. Let $\bar{p}$ be the sample mean of $\widetilde{S}_t$, and let d be the calendar day containing hour t. The fitted seasonal component is

$$\widehat{\Lambda}_t^P = \bar{p}\, \widehat{F}_d^Y\, \widehat{F}_t^D, \qquad \widehat{X}_t^P = \widetilde{S}_t - \widehat{\Lambda}_t^P. \tag{3}$$

The day-level factor $\widehat{F}_d^Y$ is estimated from daily average log-prices. If $\bar{S}_d^{\log}$ is the daily mean of $\widetilde{S}_t$, the observed factor is

$$F_d^{Y,\text{obs}} = \frac{\bar{S}_d^{\log}}{\text{yearly mean of } \bar{S}_d^{\log}}. \tag{4}$$

We regress $F_d^{Y,\text{obs}}$ on weekday and month indicators, with August split into early and late subperiods. The fitted daily values are broadcast to the hourly grid.

The intraday factor is estimated on 20 profile classes. Weekdays are grouped by month, while Saturdays and Sundays are grouped by season. Within a profile class c, the shifted log-price is regressed on hour-of-day indicators:

$$\widetilde{S}_t = a_0^{(c)} + \sum_{h=1}^{23} a_h^{(c)} \mathbf{1}_{\{\text{hour}(t)=h\}} + \varepsilon_t^{(c)}, \qquad t \in c. \tag{5}$$

The fitted intraday curve is normalized by its class mean, which gives $\widehat{F}_t^D$. This normalization makes $\widehat{F}_t^D$ a shape factor rather than a level factor.

This construction removes the main calendar, monthly, and hourly price patterns before the CARMA calibration. The remaining series $\widehat{X}^P$ is treated as the deseasonalized stochastic log-price factor.

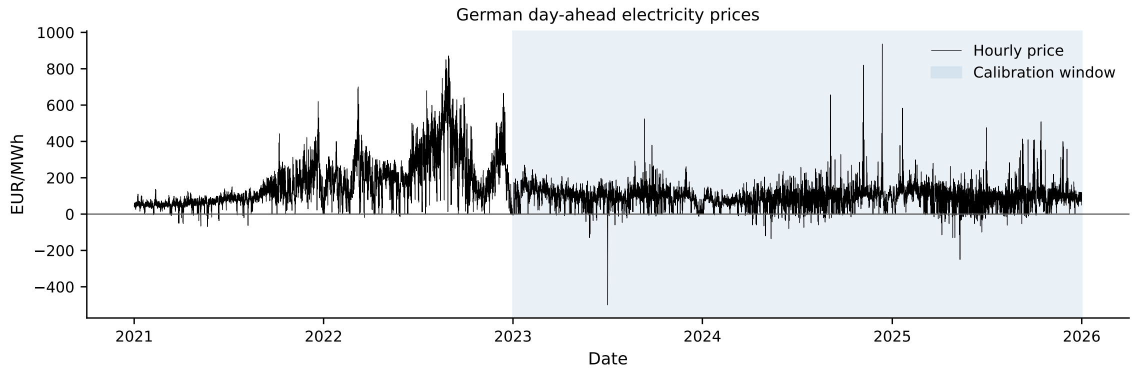

Figure 1: Hourly German day-ahead electricity prices from 2021 to 2025. The shaded region is the 2023–2025 calibration window used for the price deseasonalization and CARMA calibration.

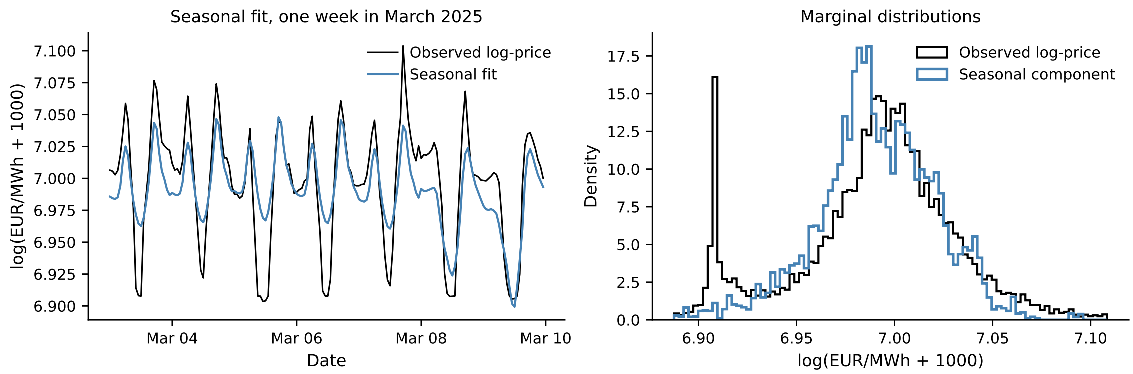

Figure 2: Price deseasonalization diagnostics showing the one-week fit of the deterministic component in shifted log-price coordinates and the marginal distributions of the observed and fitted seasonal series.

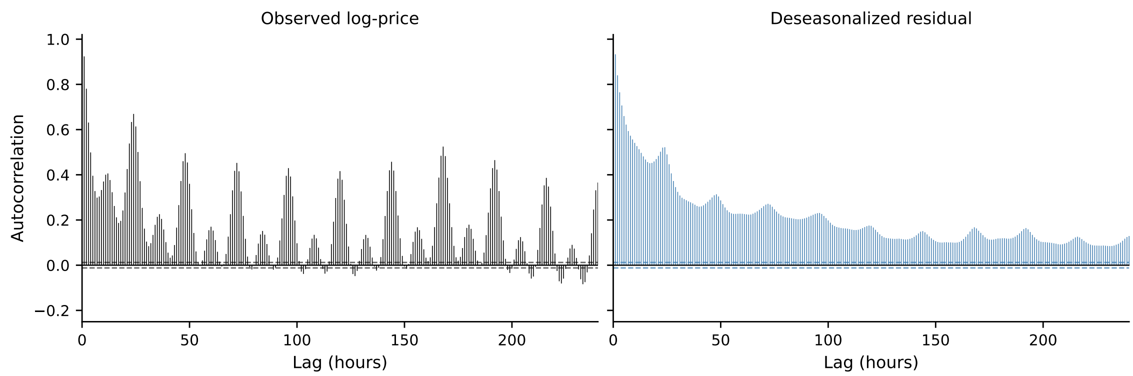

Figure 3: Empirical autocorrelation of the shifted log-price and of the deseasonalized price residual over 240 hourly lags. The deterministic deseasonalization removes the main calendar peaks, while the remaining serial dependence motivates the CARMA calibration.

2.2 Temperature

Data. We use hourly 2-meter temperature for Berlin over the same 2023–2025 UTC sample. The temperature factor is modeled in physical units, without a positivity transform:

$$T_t = \Lambda_t^T + X_t^T, \tag{6}$$

where Λ_t^T is deterministic seasonality and X_t^T is the deseasonalized stochastic temperature factor.

Deseasonalization. The deterministic component is estimated by a Fourier regression with a linear trend, a daily cycle, an annual cycle, and interactions between the two cycles. Let τ_t be the number of hours elapsed since the beginning of the sample and set $P_y = 365.25 \times 24$. The seasonal component is

$$\begin{aligned}
\Lambda_t^T(\theta) = \beta_0 + \beta_1 \tau_t &+ \sum_{k=1}^{3} \left[a_k \cos\left(\frac{2\pi k \tau_t}{24}\right) + b_k \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] \\
&+ a_y \cos\left(\frac{2\pi \tau_t}{P_y}\right) + b_y \sin\left(\frac{2\pi \tau_t}{P_y}\right) \\
&+ \sum_{k=1}^{3} \left[u_k \cos\left(\frac{2\pi k \tau_t}{24}\right) + v_k \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] \cos\left(\frac{2\pi \tau_t}{P_y}\right) \\
&+ \sum_{k=1}^{3} \left[\widetilde{u}_k \cos\left(\frac{2\pi k \tau_t}{24}\right) + \widetilde{v}_k \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] \sin\left(\frac{2\pi \tau_t}{P_y}\right).
\end{aligned} \tag{7}$$

The first sum captures the deterministic intraday pattern, the annual sine and cosine capture the low-frequency seasonal level, and the last two sums allow the intraday profile to vary smoothly over the year. All coefficients are estimated by ordinary least squares:

$$\widehat{\theta} = \arg\min_{\theta} \sum_t \left(T_t - \Lambda_t^T(\theta)\right)^2, \qquad \widehat{X}_t^T = T_t - \Lambda_t^T(\widehat{\theta}). \tag{8}$$

Figure 4 shows the fitted deterministic component over the same week used for the price diagnostics and compares the marginal distributions of observed and seasonal temperature. The ACF comparison in Figure 5 shows that the deterministic regression removes the dominant periodic component, while the residual retains serial dependence to be modeled by the temperature CARMA process.

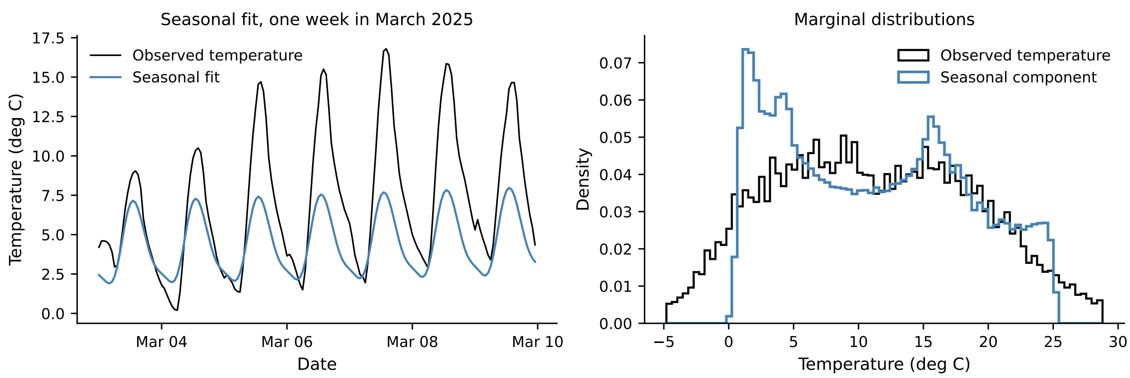

Figure 4: Temperature deseasonalization diagnostics showing the one-week fit of the deterministic component and the marginal distributions of observed and fitted seasonal temperature over the calibration sample.

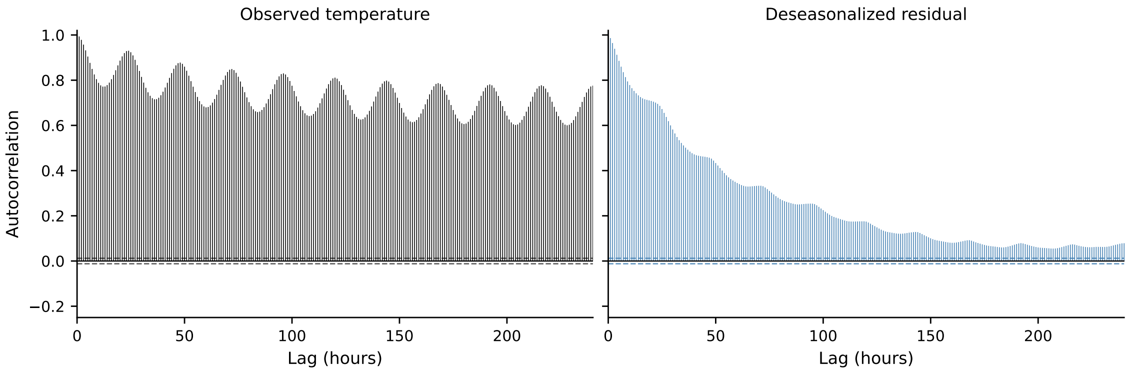

Figure 5: Empirical autocorrelation of observed temperature and of the deseasonalized temperature residual over 240 hourly lags.

2.3 Solar Capacity Factors

Data and clear-sky proxy. We use hourly German solar capacity factors $Q_t \in [0,1]$. Solar production has structural zeros at night and a daylight window that changes over the year, as shown in Figure 6. These features make a direct additive regression on Q_t inappropriate, since the seasonal level and the support of the process vary jointly over the calendar.

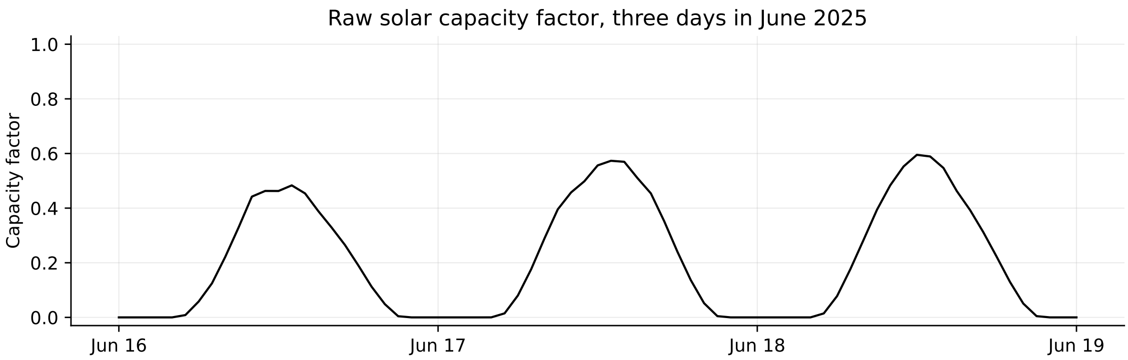

Figure 6: Raw hourly German solar capacity factor over three days in June 2025. The series combines deterministic daylight availability, cloud-driven reductions during daylight hours, and structural nighttime zeros.

We therefore first estimate an empirical clear-sky proxy. Let $d(t)$ denote the day of year and $h(t)$ the hour of day. For each calendar cell (d,h), the raw clear-sky level is defined as the upper-tail climatology

$$C^{\text{raw}}_{d,h} = q_{0.98}\{Q_t : \ d(t) = d, \ h(t) = h\}, \tag{9}$$

where $q_{0.98}$ is the empirical 98% quantile. Missing cells are filled by coarser calendar averages, and each hourly day-of-year profile is smoothed with a circular 21-day moving average. The resulting proxy is denoted

$$\widehat{C}_t = \widehat{C}_{d(t),h(t)}. \tag{10}$$

This proxy is an empirical capacity-factor envelope. It captures the seasonal and intraday availability of solar radiation while keeping the subsequent stochastic factor bounded.

Latent transformation and deseasonalization. The stochastic solar factor is defined from the shortfall of production relative to the clear-sky proxy:

$$R_t = \operatorname{clip}_{[0,1]}\left(1 - \frac{Q_t}{\max(\widehat{C}_t, \varepsilon_C)}\right), \qquad \varepsilon_C = 10^{-4}. \tag{11}$$

The bounded risk driver is then mapped to a real-valued latent coordinate by a logit transform,

$$Z_t = \mathrm{clip}_{[\varepsilon,1-\varepsilon]}\left(\frac{R_t-\alpha}{\beta}\right), \qquad Y_t^Q = \log\left(\frac{Z_t}{1-Z_t}\right), \tag{12}$$

with $\alpha = 10^{-4}$, $\beta = 0.9998$, and $\varepsilon = 10^{-4}$ in the present calibration. Deseasonalization is performed in this latent coordinate:

$$Y_t^Q = \Lambda_t^Q + X_t^Q. \tag{13}$$

The deterministic latent component is estimated by ordinary least squares. It combines hour indicators, month indicators, annual Fourier terms, and hour-by-annual interactions. With $H_h(t) = \mathbf{1}_{\{h(t)=h\}}$, $M_m(t) = \mathbf{1}_{\{\mathrm{month}(t)=m\}}$, and $\phi_t = 2\pi d(t)/365.25$, we use

$$\begin{aligned}\Lambda_t^Q(\theta) = \gamma_0 &+ \sum_{h=1}^{23}\gamma_h H_h(t) + \sum_{m=2}^{12}\eta_m M_m(t) \\ &+ \sum_{k=1}^{3}\left[a_k\cos(k\phi_t) + b_k\sin(k\phi_t)\right] \\ &+ \sum_{h=1}^{23} H_h(t)\sum_{k=1}^{2}\left[u_{h,k}\cos(k\phi_t) + v_{h,k}\sin(k\phi_t)\right].\end{aligned} \tag{14}$$

The residual is

$$\widehat{\theta} = \arg\min_{\theta}\sum_t \left(Y_t^Q - \Lambda_t^Q(\theta)\right)^2, \qquad \widehat{X}_t^Q = Y_t^Q - \Lambda_t^Q(\widehat{\theta}). \tag{15}$$

For visualization in physical units, the fitted latent seasonality is mapped back to capacity factors through

$$\widehat{Q}_t^{\mathrm{det}} = \widehat{C}_t\left[1-\alpha-\beta\sigma(\Lambda_t^Q(\widehat{\theta}))\right], \qquad \sigma(x) = \frac{1}{1+\exp(-x)}. \tag{16}$$

Figure 7 displays the main transformations over one week. The top panel shows observed production, the clear-sky envelope, and the latent seasonal component mapped back to capacity-factor scale. The middle panel shows the bounded shortfall and its normalized version. The bottom panel shows the latent coordinate Y_t^Q and its fitted seasonality.

Figure 8 gives the corresponding one-week fit and compares the marginal distributions of the latent coordinate and fitted latent deterministic component. The ACF comparison in Figure 9 shows the reduction in calendar-driven dependence after the latent deseasonalization.

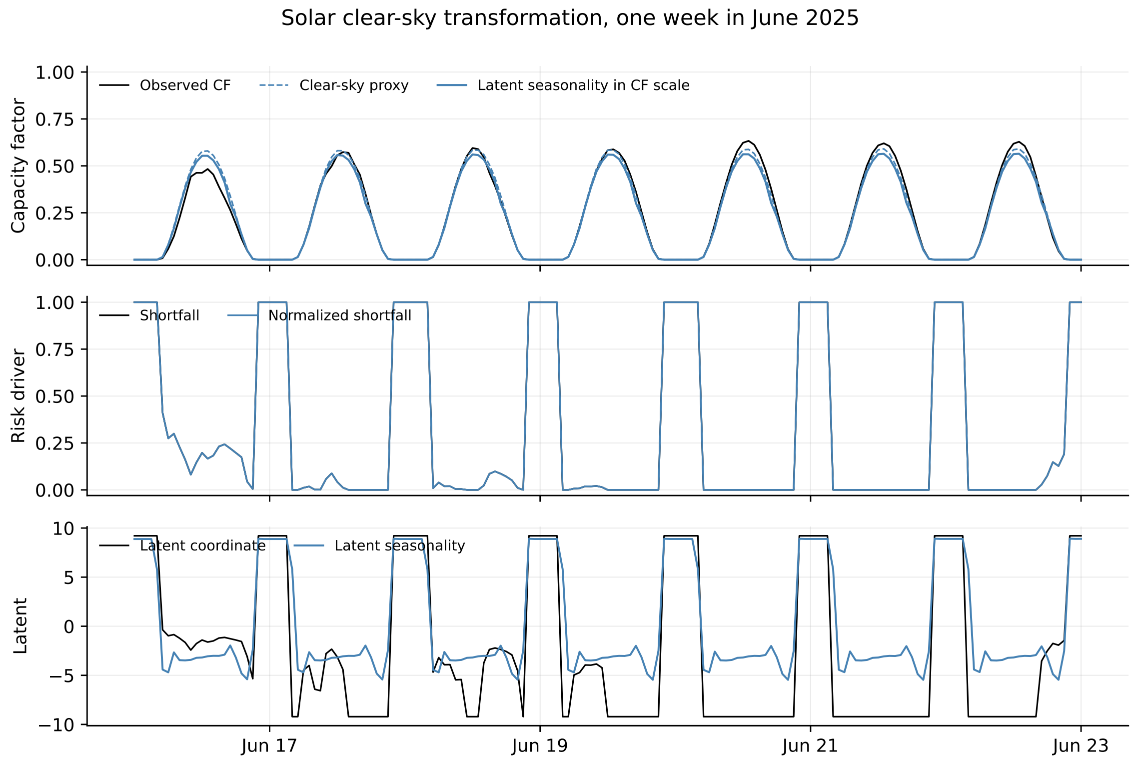

Figure 7: Solar clear-sky transformation over one week in June 2025, including the capacity-factor scale series, the shortfall variables, and the latent logit coordinate used for deseasonalization.

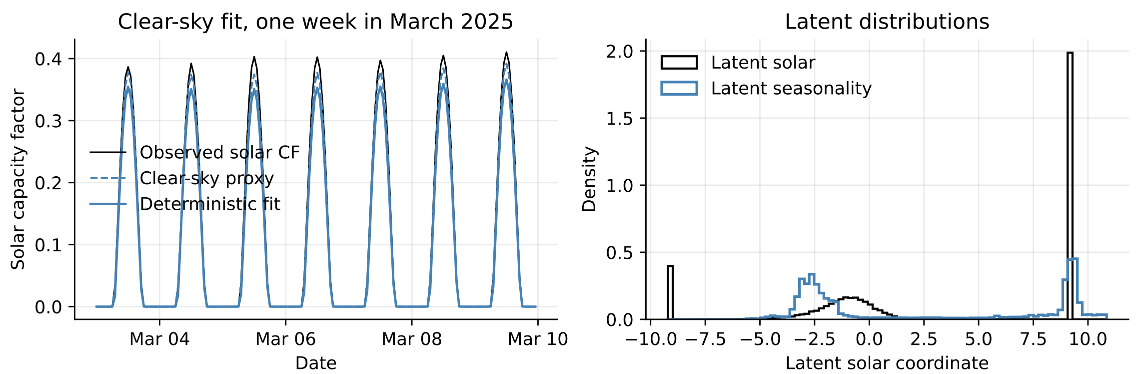

Figure 8: Solar clear-sky deseasonalization diagnostics showing the physical capacity-factor reconstruction over one week and the marginal distributions of the latent coordinate and fitted latent seasonality.

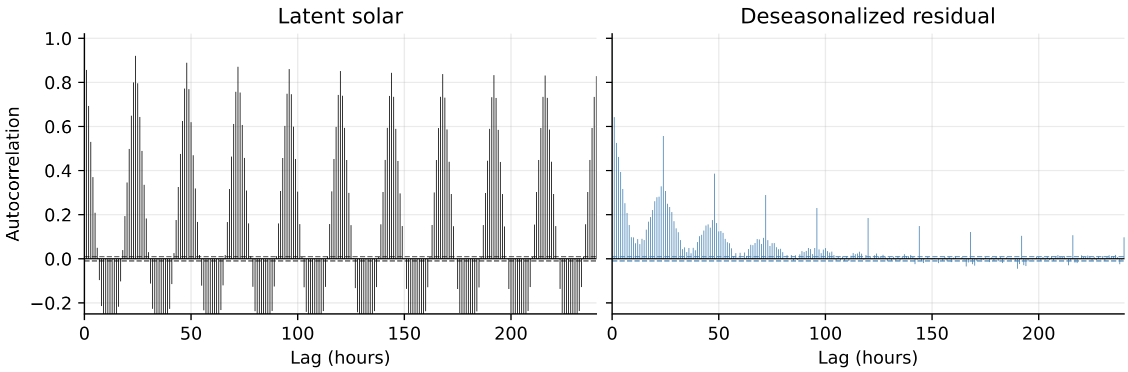

Figure 9: Empirical autocorrelation of the latent solar coordinate and of the deseasonalized solar residual over 240 hourly lags.

2.4 Wind Capacity Factors

Data and bounded transform. We use hourly German wind capacity factors $W_t \in [0, 1]$, aligned on the same UTC calendar as the German price panel. Wind does not have structural nighttime zeros, but the capacity factor is bounded and can spend long periods near low production levels. Figure 10 shows the raw series over three days. The model is therefore specified after a logit transformation of the clipped capacity factor:

$$W_t^\varepsilon = \mathrm{clip}_{[\varepsilon,1-\varepsilon]}(W_t), \qquad Y_t^W = \log\left(\frac{W_t^\varepsilon}{1 - W_t^\varepsilon}\right), \qquad \varepsilon = 10^{-4}. \tag{17}$$

The clipping only avoids infinite logit values at the boundaries. The inverse map used for physical diagnostics is

$$W_t = \sigma(Y_t^W), \qquad \sigma(x) = \frac{1}{1 + \exp(-x)}. \tag{18}$$

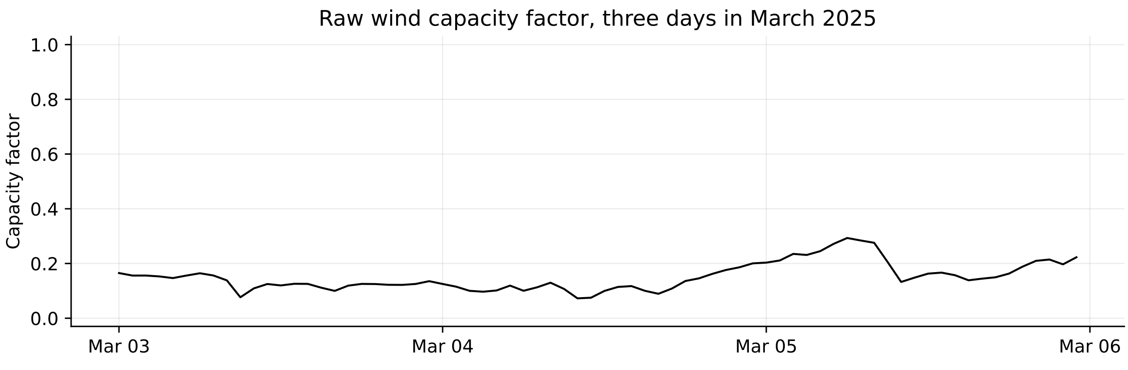

Figure 10: Raw hourly German wind capacity factor over three days in March 2025. The process is bounded, persistent, and irregular, without the deterministic day-night support constraint present in solar production.

Deseasonalization. Deseasonalization is performed in the logit coordinate:

$$Y_t^W = \Lambda_t^W + X_t^W. \tag{19}$$

The deterministic component combines intraday, weekly, monthly, and annual calendar effects. Let $H_h(t) = \mathbf{1}_{\{\mathrm{hour}(t)=h\}}$, $D_j(t) = \mathbf{1}_{\{\mathrm{weekday}(t)=j\}}$, $M_m(t) = \mathbf{1}_{\{\mathrm{month}(t)=m\}}$, and $\phi_t =$

$2\pi d(t)/365$, where $d(t)$ is the day of year. The fitted seasonality is

$$\Lambda_t^W(\theta) = \gamma_0 + \sum_{h=1}^{23} \gamma_h H_h(t) + \sum_{j=1}^{6} \delta_j D_j(t) + \sum_{m=2}^{12} \eta_m M_m(t) + \sum_{k=1}^{2} [a_k \sin(k\phi_t) + b_k \cos(k\phi_t)] . \tag{20}$$

All coefficients are estimated by ordinary least squares:

$$\widehat{\theta} = \arg\min_{\theta} \sum_t (Y_t^W - \Lambda_t^W(\theta))^2, \qquad \widehat{X}_t^W = Y_t^W - \Lambda_t^W(\widehat{\theta}). \tag{21}$$

The blue physical-scale curve in the diagnostics is not fitted directly on W_t. It is the logit seasonality mapped back as $\widehat{W}_t^{\text{sea}} = \sigma(\Lambda_t^W(\widehat{\theta}))$.

Figure 11 gives the one-week physical-scale view and compares the marginal distributions in the logit coordinate. The ACF comparison in Figure 12 shows that calendar deseasonalization removes part of the deterministic periodicity, while the remaining serial dependence is left for the wind CARMA calibration.

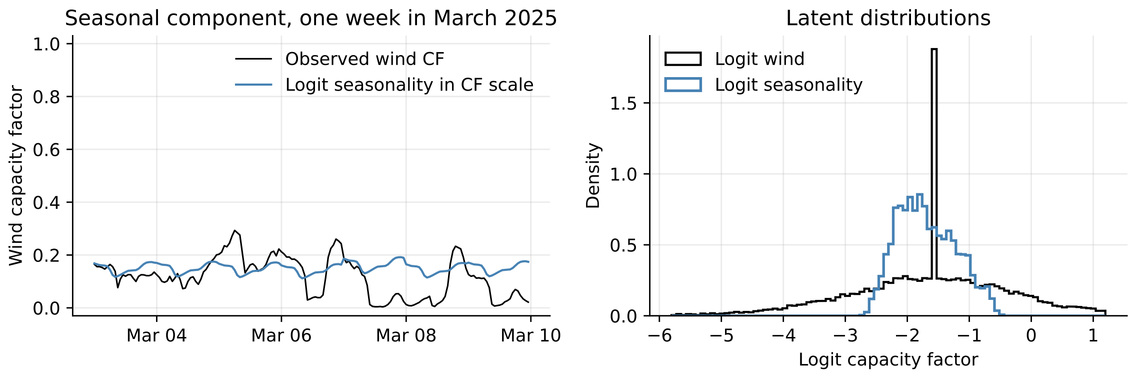

Figure 11: Wind deseasonalization diagnostics showing the one-week physical capacity-factor reconstruction from the fitted logit seasonality and the marginal distributions in the logit coordinate.

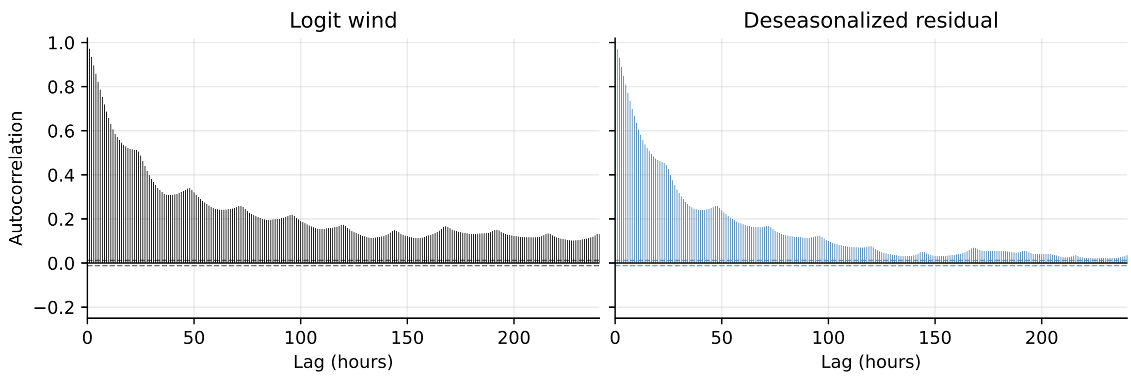

Figure 12: Empirical autocorrelation of the logit wind capacity factor and of the deseasonalized wind residual over 240 hourly lags.

3 Marginal CARMA Calibration

3.1 German Spot Prices

The deseasonalized German log-price factor $\widehat{X}^P$ is modeled by a stationary CARMA process. In state-space form,

$$dG_t = AG_t\,dt + e_p\,dL_t, \qquad X_t = b^\top G_t, \tag{22}$$

where L is a one-dimensional Levy driver, A is the companion matrix of the autoregressive polynomial

$$a(z) = z^p + a_1 z^{p-1} + \cdots + a_p, \tag{23}$$

and $b = (b_0, \ldots, b_q, 0, \ldots, 0)^\top$ contains the moving-average coefficients. Stability requires the roots of a to have negative real parts. For a fixed AR structure, the theoretical autocorrelation is a linear combination of exponentials generated by these roots. Real roots generate monotone exponential decay, while a complex conjugate pair generates a damped oscillation.

ACF-based multiscale initialization. The CARMA order is selected from the empirical autocorrelation of $\widehat{X}^P$. The retained ACF model is

$$\rho_{\mathrm{ms}}(h) = \sum_{i=1}^{3} w_i \exp(-\kappa_i h) + w_4 \exp(-\kappa_4 h)\cos\left(\frac{2\pi h}{24}\right), \qquad \sum_{i=1}^{4} w_i = 1, \quad w_i \geq 0, \tag{24}$$

with h measured in hours. The cosine frequency is fixed at $2\pi/24$, since the empirical ACF exhibits a clear 24-hour oscillatory component. The three real modes represent different persistence scales: a fast component of a few hours, an intermediate component of about one day, and a slow component. The slow half-life is fixed at 120 days at this stage in order to impose a long-memory component needed for option pricing horizons. This constraint is used only to build a stable and interpretable initial CARMA structure; the slow real half-life is re-estimated in the QMLE step.

The fitted multiscale decomposition has three real roots and one complex conjugate pair. Therefore the autoregressive order is

$$p = 3 + 2 = 5.$$

We use the canonical full moving-average order $q = p - 1 = 4$, and the spectral factorization of the fitted ACF gives a stable and invertible initial CARMA(5,4) model.

Table 1: German price CARMA(5,4) parameters from the ACF initialization.

Quantity	Value	HL h	HL d	Status
Log-likelihood	77561.686			before QMLE
Levy drift m	1.396680×10^{-5}			profiled
Levy variance ν^2	1.879996×10^{-4}			profiled
λ_f fast real	-0.263460	2.63	0.110	free in ACF fit
λ_m intermediate real	-0.024161	28.69	1.195	free in ACF fit
λ_s slow real	-2.4068×10^{-4}	2880.00	120.00	fixed in ACF fit
$\lambda_d^{\pm}$ daily pair	$-0.004004 \pm 0.261799i$	173.11	7.213	period fixed
Daily period	24.00 h			fixed
a_1	2.958700×10^{-1}			implied by roots
a_2	7.729482×10^{-2}			implied by roots
a_3	1.978743×10^{-2}			implied by roots
a_4	4.411413×10^{-4}			implied by roots
a_5	1.050270×10^{-7}			implied by roots
b_0	7.835423×10^{-6}			spectral factorization
b_1	7.187126×10^{-3}			spectral factorization
b_2	7.131652×10^{-2}			spectral factorization
b_3	1.304485×10^{-1}			spectral factorization
b_4	1.000000			normalization

Table 2: German price CARMA(5,4) parameters after joint QMLE.

Quantity	Value	HL h	HL d	Status
Log-likelihood	77709.048			after QMLE
Levy drift m	8.473621×10^{-6}			profiled
Levy variance ν^2	1.753615×10^{-4}			profiled
λ_f fast real	-0.153231	4.52	0.188	re-estimated
λ_m intermediate real	-0.019289	35.94	1.497	re-estimated
λ_s slow real	-3.5066×10^{-4}	1976.67	82.36	re-estimated
$\lambda_d^{\pm}$ daily pair	$-0.004004 \pm 0.261799i$	173.11	7.213	fixed from ACF step
Daily period	24.00 h			fixed
a_1	1.808780×10^{-1}			implied by roots
a_2	7.295539×10^{-2}			implied by roots
a_3	1.187629×10^{-2}			implied by roots
a_4	2.067768×10^{-4}			implied by roots
a_5	7.105209×10^{-8}			implied by roots
b_0	4.892285×10^{-6}			re-estimated
b_1	3.871538×10^{-3}			re-estimated
b_2	6.724760×10^{-2}			re-estimated
b_3	8.086069×10^{-2}			re-estimated
b_4	1.000000			normalization

The fixed daily pair fixes its quadratic AR factor. The reported coefficients $a_1, \ldots, a_5$ are nevertheless the coefficients of the full AR polynomial, obtained after multiplying this fixed factor by the three real-root factors.

Figure 13 compares the empirical ACF with the multiscale initialization and with the final QMLE CARMA ACF over 700 hourly lags. The initialization captures both the fast decay and the persistent daily oscillations. The QMLE step preserves this structure while adapting the real half-lives and the moving-average polynomial to the exact likelihood. Compared with the weather and renewable marginals below, the price model is the only one that keeps three real modes in addition to the daily pair. The extra slow mode is not introduced to improve short-lag fit; it carries persistence at option-pricing horizons, where a few-day CARMA memory would be too short.

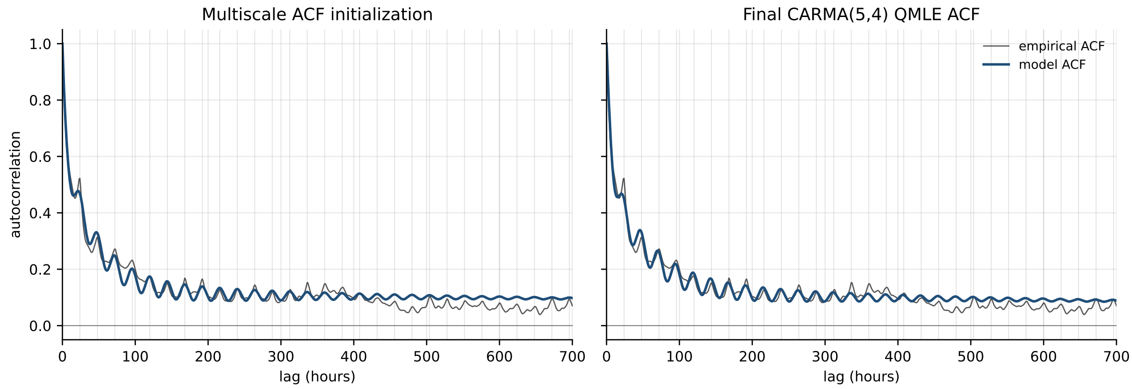

Figure 13: German price ACF calibration over 700 hourly lags, comparing the empirical autocorrelation with the multiscale initialization and with the final CARMA(5,4) autocorrelation after QMLE.

Exact Gaussian QMLE. Starting from the ACF-based CARMA(5,4), the final marginal parameters are estimated by exact Gaussian prediction-error QMLE. For given CARMA coefficients, the Kalman recursion gives one-step prediction errors e_i and innovation variances r_i under unit Levy variance. Since the Levy drift enters as a constant mean shift, the adjusted innovation can be written

$$\varepsilon_i(m) = e_i - mc_i.$$

The drift rate is profiled by weighted least squares,

$$\widehat{m} = \frac{\sum_i e_i c_i / r_i}{\sum_i c_i^2 / r_i}, \qquad \widehat{\nu}^2 = \frac{1}{n} \sum_i \frac{\varepsilon_i(\widehat{m})^2}{r_i}. \tag{25}$$

The numerical objective is the reduced Gaussian prediction-error likelihood augmented by a soft ACF penalty,

$$\mathcal{J}(\theta) = \log(\widehat{\nu}^2) + \frac{1}{n} \sum_i \log r_i + \lambda_{\mathrm{acf}} \frac{1}{|H|} \sum_{h \in H} \omega_h \left(\rho_\theta(h) - \widehat{\rho}(h)\right)^2. \tag{26}$$

The ACF penalty stabilizes the long-lag dependence without replacing the likelihood criterion. In the German price calibration, $\lambda_{\mathrm{acf}} = 10$, the final log-likelihood is 77709.05, and the profiled Levy variance rate is $\widehat{\nu}^2 = 1.7536 \times 10^{-4}$.

The ARMA likelihoods are reported only as empirical benchmarks on the same hourly residual series. They are not directly comparable as pricing models, since they do not define the continuous-time CARMA dynamics used below for simulation and option valuation.

Table 3: Discrete-time ARMA log-likelihood benchmarks on the same deseasonalized German log-price residuals. These benchmarks are not used for pricing.

Model	Estimation	Log-likelihood	Comment
ARMA(1,0)	discrete-time ML	77454.051	stationary benchmark
ARMA(5,4)	discrete-time state-space ML	78660.514	LBFGS convergence flag not satisfied

Driver distribution. After the CARMA parameters are fixed, hourly increments of the latent Levy driver are recovered from the smoothed state. We fit both a Gaussian law and a normal inverse Gaussian law to these increments. The fitted parameters and moments are reported in Table 4.

Table 4: Fitted moments for recovered hourly Levy increments in the German price CARMA model.

Distribution	Mean	Std. dev.	Skewness	Excess kurtosis
Gaussian	1.8906×10^{-6}	1.3236×10^{-2}	0.000	0.000
NIG	1.8879×10^{-6}	1.2135×10^{-2}	-0.563	9.675

Table 5: Distribution parameters for the recovered German price driver.

Distribution	Parameter	Value
Gaussian	μ	1.8906×10^{-6}
Gaussian	σ	1.3236×10^{-2}
NIG	α	47.467
NIG	β	-5.074
NIG	δ	6.870×10^{-3}
NIG	μ	7.405×10^{-4}

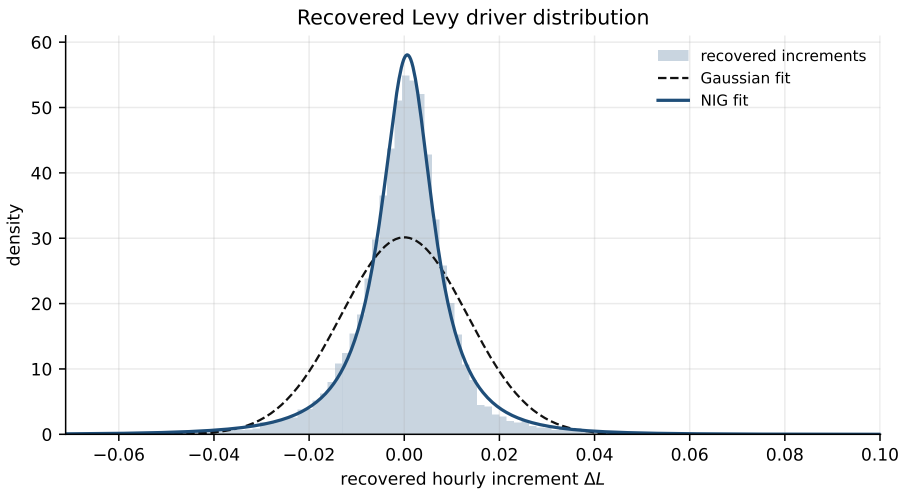

Figure 14: Recovered hourly Levy driver increments for the German price CARMA model. The Gaussian fit underestimates the concentration around zero and the tail thickness, while the NIG fit gives the retained marginal driver law.

Figure 15 shows the effect of this driver law at three levels: the deseasonalized log-price residual, the final shifted log-price, and the final price in EUR/MWh. The same seasonal component is used when mapping simulated CARMA residuals back to log-prices and prices.

The CARMA residual model improves the continuous stochastic component but does not reproduce the bimodal empirical distribution of final prices, in particular the concentration

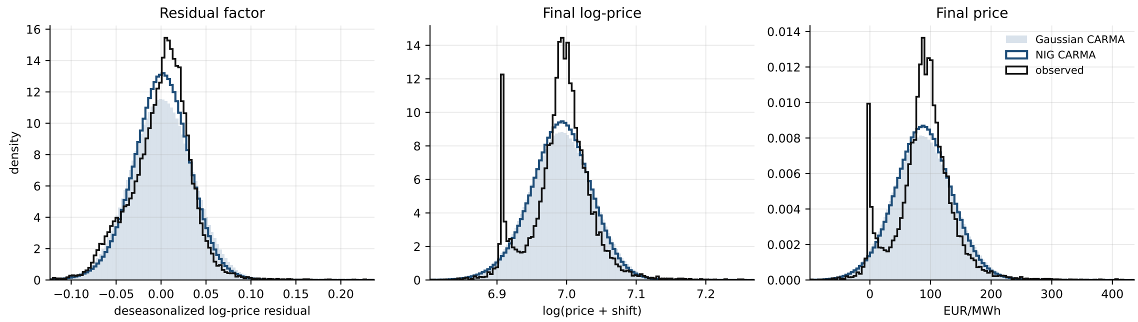

Figure 15: Distribution checks for the German price marginal model at the residual factor, shifted log-price, and EUR/MWh price levels.

around zero EUR/MWh. That feature is induced by the market price formation mechanism and is not captured by a smooth continuous-time CARMA marginal factor alone. The price driver is heavy-tailed, but its excess kurtosis is lower than for solar and wind. The remaining mismatch therefore comes less from the Levy driver alone than from the discontinuous empirical shape of final power prices.

Pure QMLE comparison. We also ran a pure exact QMLE without the ACF penalty ($\lambda_{\text{acf}} = 0$), using semi-global restarts from the same CARMA(5,4) structure. The best run over 20 restarts increases the Gaussian log-likelihood because the objective only rewards one-step prediction fit; it does not penalize the loss of long-lag ACF structure.

Table 6: Penalized versus pure QMLE for the German price CARMA(5,4).

Quantity	Retained QMLE	Pure QMLE
ACF penalty λ_{acf}	10	0
Log-likelihood	77709.05	78808.20
Fast real half-life	4.52 h	0.51 h
Intermediate real half-life	35.94 h	1.16 h
Slow real half-life	82.36 d	1.36 d
Daily-pair half-life	7.21 d	5.21 d
Daily period	24.00 h	24.00 h

The pure QMLE is therefore better as a one-step Gaussian likelihood fit, but it collapses the slow price factor from several months to about one day. This is not appropriate for the pricing step, because the model would lose the long-horizon persistence imposed by the ACF initialization and retained by the penalized QMLE.

3.2 German Temperature

The stochastic temperature factor $\widehat{X}^T$ is the hourly Berlin temperature residual after deterministic Fourier deseasonalization. It is modeled with the same stationary CARMA state-space structure as in eq. (22). The empirical ACF is smoother than the electricity price ACF, but still contains a daily oscillatory component. The initial ACF model is therefore

$$\rho^T_{\text{ms}}(h) = w_f e^{-\kappa_f h} + w_s e^{-\kappa_s h} + w_d e^{-\kappa_d h} \cos\left(\frac{2\pi h}{24}\right), \qquad w_f + w_s + w_d = 1, \quad w_f, w_s, w_d \geq 0. \tag{27}$$

The two real components represent a relatively fast decay and a slower persistent component. The complex conjugate pair enforces the 24-hour oscillation. This gives two real roots and one complex pair, hence

$$p = 2 + 2 = 4, \qquad q = p - 1 = 3.$$

The daily frequency is fixed in the ACF step. In the subsequent QMLE step, the two real AR roots and the moving-average polynomial are re-estimated, while the daily AR pair is kept fixed to preserve the interpreted oscillatory component.

Table 7: German temperature CARMA(4,3) parameters from the ACF initialization.

Quantity	Value	HL h	HL d	Status
Log-likelihood	-23759.581			before QMLE
Levy drift m	5.479170×10^{-4}			profiled
Levy variance ν^2	3.497107×10^{-1}			profiled
λ_f fast real	-0.027210	25.47	1.061	free in ACF fit
λ_s slow real	-0.011917	58.17	2.424	free in ACF fit
$\lambda_d^{\pm}$ daily pair	$-0.009532 \pm 0.261799i$	72.72	3.030	period fixed
Daily period	24.00 h			fixed
a_1	5.818961×10^{-2}			implied by roots
a_2	6.969989×10^{-2}			implied by roots
a_3	2.691419×10^{-3}			implied by roots
a_4	2.225311×10^{-5}			implied by roots
b_0	1.363967×10^{-3}			spectral factorization
b_1	6.904682×10^{-2}			spectral factorization
b_2	7.815313×10^{-2}			spectral factorization
b_3	1.000000			normalization

Table 8: German temperature CARMA(4,3) parameters after joint QMLE.

Quantity	Value	HL h	HL d	Status
Log-likelihood	-22940.645			after QMLE
Levy drift m	2.853529×10^{-4}			profiled
Levy variance ν^2	1.606579×10^{-1}			profiled
λ_f fast real	-1.308587	0.53	0.022	re-estimated
λ_s slow real	-0.018198	38.09	1.587	re-estimated
$\lambda_d^{\pm}$ daily pair	$-0.009532 \pm 0.261799i$	72.72	3.030	fixed from ACF step
Daily period	24.00 h			fixed
a_1	1.345849			implied by roots
a_2	1.177365×10^{-1}			implied by roots
a_3	9.151094×10^{-2}			implied by roots
a_4	1.634346×10^{-3}			implied by roots
b_0	1.633022×10^{-1}			re-estimated
b_1	2.010868×10^{-1}			re-estimated
b_2	2.567675			re-estimated
b_3	1.000000			normalization

The fixed daily AR pair fixes its quadratic factor. The full AR coefficients $a_1, \ldots, a_4$ are reported after multiplying this fixed factor by the two real-root factors, so they change when the real roots are re-estimated.

Figure 16 compares the empirical temperature ACF with the multiscale initialization and the final QMLE CARMA ACF over 700 hourly lags. The QMLE step shortens the fast real scale while preserving the daily oscillatory structure and the slower decay. After QMLE, the slow real half-life is about one and a half days, close to the solar value and much shorter than the wind and price slow modes. Temperature is therefore the cleanest two-scale physical marginal: a very fast adjustment, a one-to-two-day persistence scale, and the fixed daily oscillation.

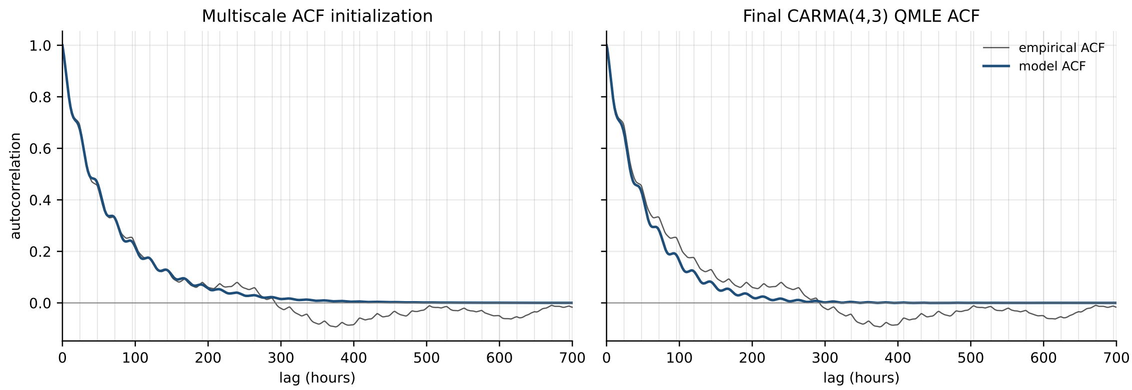

Figure 16: German temperature ACF calibration over 700 hourly lags, comparing the empirical autocorrelation with the multiscale initialization and with the final CARMA(4,3) autocorrelation after QMLE.

Exact Gaussian QMLE and benchmarks. The same exact Gaussian prediction-error likelihood as in eq. (25) is used for the temperature residual. The ACF penalty is smaller than for prices, with $\lambda_{\text{acf}} = 0.1$. The final log-likelihood is -22940.65, and the profiled Levy variance rate is $\widehat{\nu}^2 = 1.6066 \times 10^{-1}$.

Table 9: Discrete-time ARMA log-likelihood benchmarks on the same deseasonalized German temperature residuals. These benchmarks are not used for pricing.

Model	Estimation	Log-likelihood
ARMA(1,0)	discrete-time ML	-24611.865
ARMA(4,3)	discrete-time state-space ML	-23173.448

Driver distribution. Hourly increments of the recovered temperature Levy driver are fitted with both a Gaussian law and a normal inverse Gaussian law. The NIG law is retained because it captures the negative skewness and excess kurtosis of the recovered increments.

Table 10: Fitted moments for recovered hourly Levy increments in the German temperature CARMA model.

Distribution	Mean	Std. dev.	Skewness	Excess kurtosis
Gaussian	-2.5832×10^{-4}	3.9309×10^{-1}	0.000	0.000
NIG	-2.5832×10^{-4}	3.8501×10^{-1}	-0.477	5.205

Table 11: Distribution parameters for the recovered German temperature driver.

Distribution	Parameter	Value
Gaussian	μ	-2.5832×10^{-4}
Gaussian	σ	3.9309×10^{-1}
NIG	α	2.064
NIG	β	-0.257
NIG	δ	2.9888×10^{-1}
NIG	μ	3.7243×10^{-2}

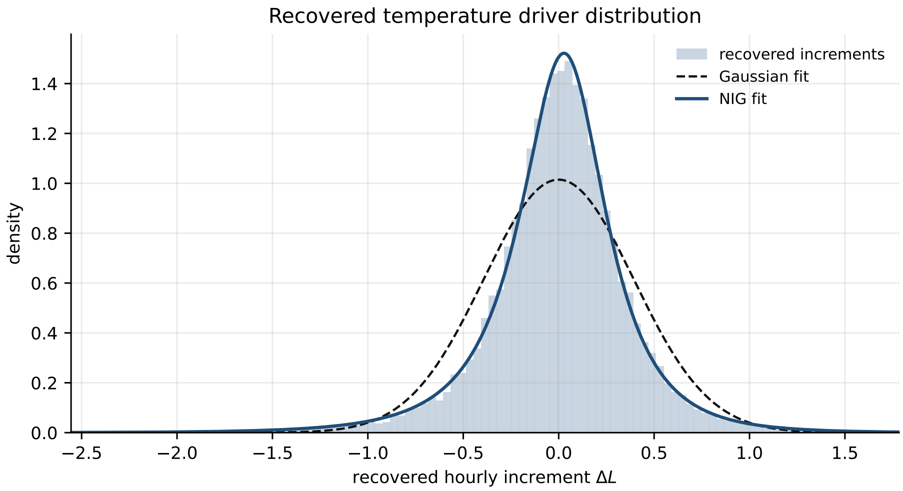

Figure 17: Recovered hourly Levy driver increments for the German temperature CARMA model. The NIG fit captures the asymmetric and heavy-tailed driver distribution better than the Gaussian fit.

Figure 18 reports the induced distributions at the residual, residual-increment, and physical temperature levels. The same Fourier seasonality is added back when simulated residuals are mapped to degrees Celsius. Unlike solar and wind, this final mapping is additive and unbounded. As a result, distributional errors in the CARMA residual are not amplified or compressed by a logit-type inverse transform.

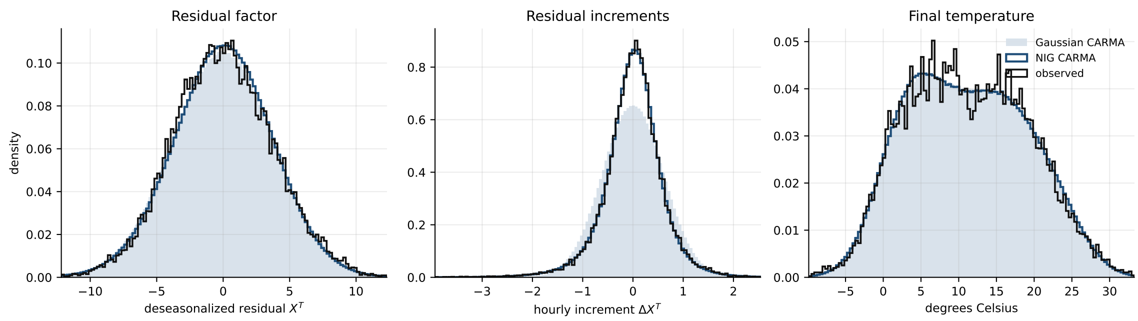

Figure 18: Distribution checks for the German temperature marginal model at the deseasonalized residual, hourly residual-increment, and physical temperature levels.

3.3 German Solar Production

The solar marginal model is fitted to the deseasonalized latent residual $\widehat{X}^Q$ defined after the clear-sky normalization and logit transformation in section 2.3. The physical capacity factor is bounded and has structural nighttime zeros, so the CARMA model is not applied directly to Q_t. It is applied to the transformed residual, and simulated paths are mapped back through the inverse clear-sky transformation.

The empirical ACF of $\widehat{X}^Q$ has a sharp short-lag decay, a slower component, and a residual daily oscillation. The ACF initialization therefore uses the same two-real-plus-one-complex structure as temperature:

$$\rho^Q_{\mathrm{ms}}(h) = w_f e^{-\kappa_f h} + w_s e^{-\kappa_s h} + w_d e^{-\kappa_d h} \cos\left(\frac{2\pi h}{24}\right), \qquad w_f + w_s + w_d = 1, \quad w_f, w_s, w_d \geq 0. \tag{28}$$

The fast real component captures sub-hour to intraday adjustment in the latent cloudiness factor, the slower real component captures persistence over roughly one to two days, and the complex pair imposes the 24-hour oscillatory mode. This gives a CARMA(4,3) specification.

Table 12: German solar CARMA(4,3) parameters from the ACF initialization.

Quantity	Value	HL h	HL d	Status
Log-likelihood	-84636.518			before QMLE
Levy drift m	1.136470×10^{-4}			profiled
Levy variance ν^2	9.270238			profiled
λ_f fast real	-1.370835	0.51	0.021	free in ACF fit
λ_s slow real	-0.016625	41.69	1.737	free in ACF fit
$\lambda_d^{\pm}$ daily pair	$-0.009845 \pm 0.261799i$	70.41	2.934	period fixed
Daily period	24.00 h			fixed
a_1	1.407149			implied by roots
a_2	1.187446×10^{-1}			implied by roots
a_3	9.567819×10^{-2}			implied by roots
a_4	1.564222×10^{-3}			implied by roots
b_0	8.477053×10^{-3}			spectral factorization
b_1	8.361212×10^{-2}			spectral factorization
b_2	2.305833×10^{-1}			spectral factorization
b_3	1.000000			normalization

Table 13: German solar CARMA(4,3) parameters after joint QMLE.

Quantity	Value	HL h	HL d	Status
Log-likelihood	-84622.936			after QMLE
Levy drift m	1.269139×10^{-4}			profiled
Levy variance ν^2	9.122430			profiled
λ_f fast real	-1.353919	0.51	0.021	re-estimated
λ_s slow real	-0.018410	37.65	1.569	re-estimated
$\lambda_d^{\pm}$ daily pair	$-0.009845 \pm 0.261799i$	70.41	2.934	fixed from ACF step
Daily period	24.00 h			fixed
a_1	1.392018			implied by roots
a_2	1.205815×10^{-1}			implied by roots
a_3	9.468167×10^{-2}			implied by roots
a_4	1.710750×10^{-3}			implied by roots
b_0	8.992189×10^{-3}			re-estimated
b_1	8.187720×10^{-2}			re-estimated
b_2	2.392272×10^{-1}			re-estimated
b_3	1.000000			normalization

The daily AR pair is fixed after the ACF step. The QMLE step therefore adjusts the two real persistence scales and the MA polynomial while preserving the interpreted 24-hour oscillation.

Figure 19 compares the empirical ACF with the multiscale initialization and the final QMLE CARMA ACF over 700 hourly lags. The final fit keeps the short-lag decay and the daily oscillatory structure, while the remaining long-lag empirical dependence is only partially captured by this four-dimensional marginal factor. The solar real half-lives are close to the temperature half-lives after QMLE, but their interpretation is different. The fast solar mode reflects cloudiness in the clear-sky coordinate, while the daily component is tied to the production support itself; wind below instead puts more weight on slow weather-regime persistence.

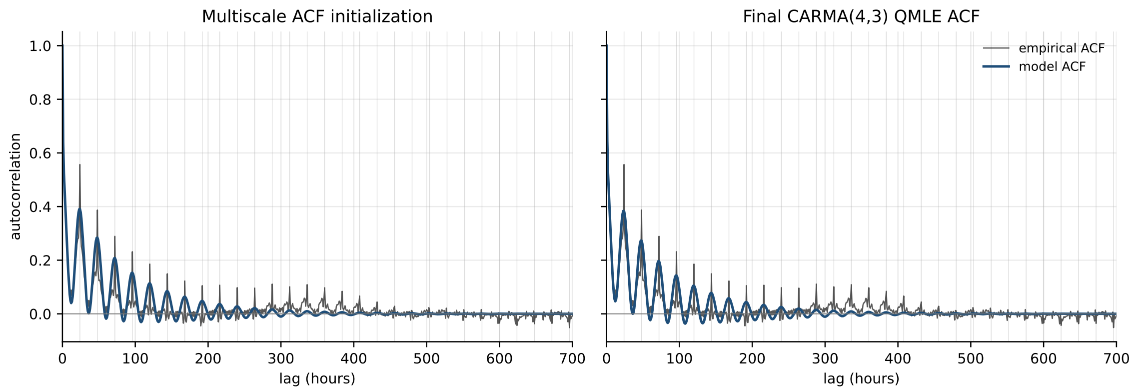

Figure 19: German solar latent-residual ACF calibration over 700 hourly lags, comparing the empirical autocorrelation with the multiscale initialization and with the final CARMA(4,3) autocorrelation after QMLE.

Exact Gaussian QMLE and benchmarks. The exact Gaussian prediction-error likelihood is applied to the latent residual $\widehat{X}^Q$. The ACF penalty is set to $\lambda_{\text{acf}} = 1000$, since the solar residual ACF contains a strong daily component that must be preserved by the continuous-time marginal model. The final log-likelihood is -84622.94, and the profiled Levy variance rate is $\widehat{\nu}^2 = 9.1224$.

Table 14: Discrete-time ARMA log-likelihood benchmarks on the same deseasonalized German solar latent residuals. These benchmarks are not used for pricing.

Model	Estimation	Log-likelihood
ARMA(1,0)	discrete-time ML	-86462.196
ARMA(4,3)	discrete-time state-space ML	-84785.671

Driver distribution. The recovered solar Levy increments are strongly non-Gaussian. The NIG law is therefore retained because it captures the negative skewness and high excess kurtosis that are absent from a Gaussian driver. Among the four marginals, solar has the largest negative skewness and excess kurtosis in the recovered driver. This is consistent with abrupt cloud-driven changes in the latent coordinate and with the bounded physical output after the inverse clear-sky transformation.

Table 15: Fitted moments for recovered hourly Levy increments in the German solar CARMA model.

Distribution	Mean	Std. dev.	Skewness	Excess kurtosis
Gaussian	1.3308×10^{-4}	2.8964	0.000	0.000
NIG	1.3310×10^{-4}	3.0420	-0.803	18.789

Table 16: Distribution parameters for the recovered German solar driver.

Distribution	Parameter	Value
Gaussian	μ	1.3308×10^{-4}
Gaussian	σ	2.8964
NIG	α	1.3609×10^{-1}
NIG	β	-1.4892×10^{-2}
NIG	δ	1.2368
NIG	μ	1.3629×10^{-1}

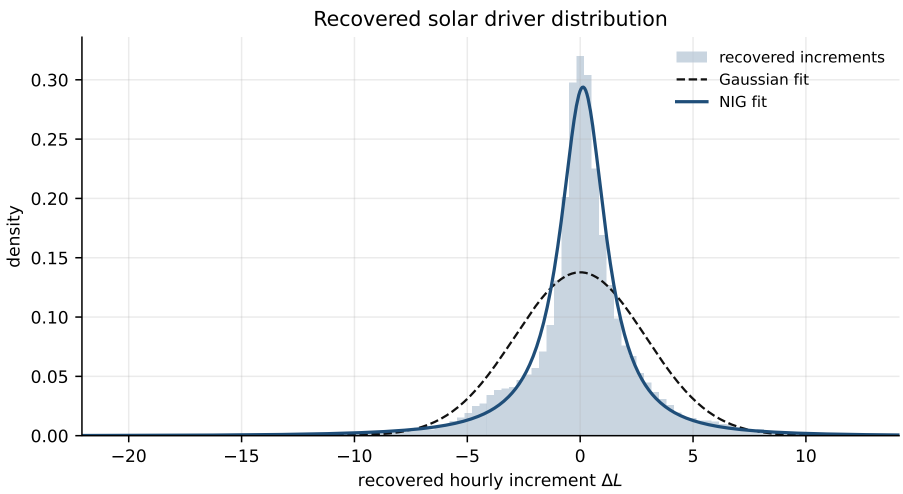

Figure 20: Recovered hourly Levy driver increments for the German solar CARMA model. The NIG fit captures the asymmetric and heavy-tailed driver distribution better than the Gaussian fit.

Figure 21 reports the induced distributions at the latent residual, residual-increment, and physical capacity-factor levels. The physical mapping uses the same clear-sky envelope, fitted transform bounds, and latent seasonal component as the deseasonalization step.

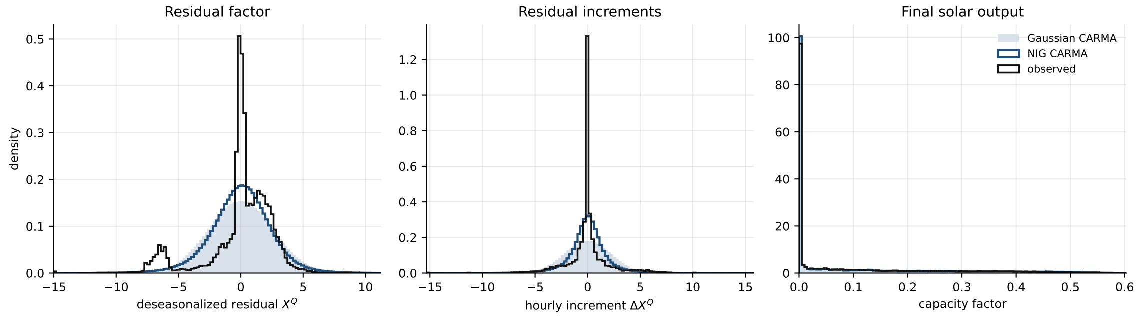

Figure 21: Distribution checks for the German solar marginal model at the deseasonalized latent residual, hourly residual-increment, and physical capacity-factor levels.

The CARMA-NIG model improves the latent residual tails relative to a Gaussian driver, but it does not fully reproduce the left-skewed empirical residual law. Part of this asymmetry comes from the bounded clear-sky transformation and the structural zero-production periods in physical solar output.

3.4 German Wind Production

The wind marginal model is fitted to the deseasonalized logit residual $\widehat{X}^W$ defined in section 2.4. As for solar, the CARMA model is estimated in an unconstrained latent coordinate and simulated paths are mapped back to capacity factors with the inverse logit transform. This keeps the continuous-time model away from the physical bounds 0 and 1.

The empirical wind ACF is less dominated by a deterministic day-night structure than the solar ACF, but it still contains a weak daily oscillatory component and several persistence scales. The ACF initialization uses

$$\rho^W_{\mathrm{ms}}(h) = w_f e^{-\kappa_f h} + w_s e^{-\kappa_s h} + w_d e^{-\kappa_d h} \cos\left(\frac{2\pi h}{24}\right), \qquad w_f + w_s + w_d = 1, \quad w_f, w_s, w_d \geq 0. \tag{29}$$

The first real mode captures intraday persistence, the second real mode captures slower weather-regime persistence, and the complex pair fixes the 24-hour oscillation. This again gives a CARMA(4,3) specification.

Table 17: German wind CARMA(4,3) parameters from the ACF initialization.

Quantity	Value	HL h	HL d	Status
Log-likelihood	-7013.176			before QMLE
Levy drift m	6.357371×10^{-5}			profiled
Levy variance ν^2	1.015442×10^{-1}			profiled
λ_f fast real	-0.043126	16.07	0.670	free in ACF fit
λ_s slow real	-0.003883	178.51	7.438	free in ACF fit
$\lambda_d^{\pm}$ daily pair	$-0.008797 \pm 0.261799i$	78.80	3.283	period fixed
Daily period	24.00 h			fixed
a_1	6.460237×10^{-2}			implied by roots
a_2	6.961081×10^{-2}			implied by roots
a_3	3.228522×10^{-3}			implied by roots
a_4	1.149037×10^{-5}			implied by roots
b_0	4.273146×10^{-4}			spectral factorization
b_1	6.854653×10^{-2}			spectral factorization
b_2	4.721584×10^{-2}			spectral factorization
b_3	1.000000			normalization

Table 18: German wind CARMA(4,3) parameters after joint QMLE.

Quantity	Value	HL h	HL d	Status
Log-likelihood	-6983.728			after QMLE
Levy drift m	1.572834×10^{-5}			profiled
Levy variance ν^2	9.934486×10^{-2}			profiled
λ_f fast real	-0.033136	20.92	0.872	re-estimated
λ_s slow real	-7.6809×10^{-4}	902.44	37.60	re-estimated
$\lambda_d^{\pm}$ daily pair	$-0.008797 \pm 0.261799i$	78.80	3.283	fixed from ACF step
Daily period	24.00 h			fixed
a_1	5.149728×10^{-2}			implied by roots
a_2	6.923824×10^{-2}			implied by roots
a_3	2.326800×10^{-3}			implied by roots
a_4	1.746357×10^{-6}			implied by roots
b_0	9.228613×10^{-5}			re-estimated
b_1	6.888442×10^{-2}			re-estimated
b_2	5.391827×10^{-2}			re-estimated
b_3	1.000000			normalization

The QMLE step increases the slow real half-life to about 38 days. This mode is not a calendar effect; it represents persistent wind-regime variation in the latent residual. The daily AR pair remains fixed from the ACF step. This is the main contrast with solar. Both renewable marginals use a bounded latent transform, but wind keeps a much longer real persistence scale, whereas solar is dominated by short-lag cloud effects and the deterministic day-night structure.

Figure 22 compares the empirical ACF with the multiscale initialization and the final QMLE CARMA ACF over 700 hourly lags. The fit captures the main short-lag decay and the broad persistent component, while the remaining small oscillations are absorbed through the fixed daily pair.

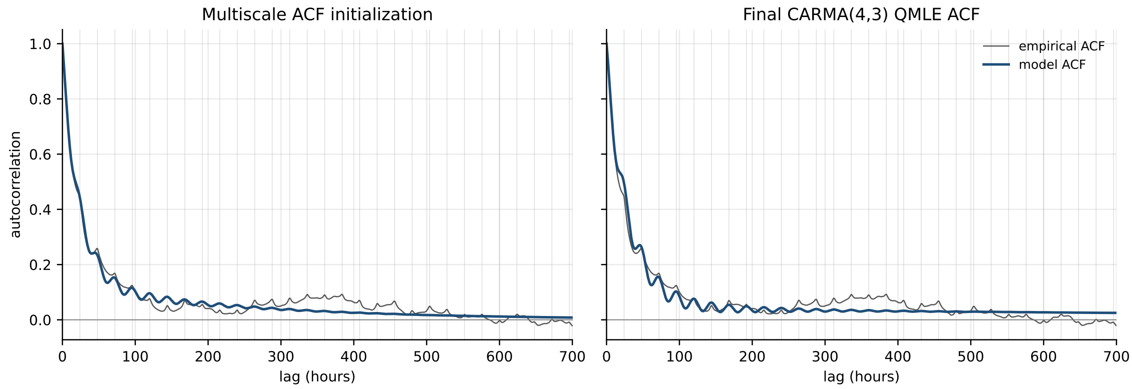

Figure 22: German wind logit-residual ACF calibration over 700 hourly lags, comparing the empirical autocorrelation with the multiscale initialization and with the final CARMA(4,3) autocorrelation after QMLE.

Exact Gaussian QMLE and benchmarks. The exact Gaussian prediction-error likelihood is applied to the wind logit residual $\widehat{X}^W$, with $\lambda_{\mathrm{acf}} = 0.1$. The final log-likelihood is -6983.73, and the profiled Levy variance rate is $\widehat{\nu}^2 = 9.9345 \times 10^{-2}$.

Table 19: Discrete-time ARMA log-likelihood benchmarks on the same deseasonalized German wind logit residuals. These benchmarks are not used for pricing.

Model	Estimation	Log-likelihood
ARMA(1,0)	discrete-time ML	-7261.653
ARMA(4,3)	discrete-time state-space ML	-6853.455

Driver distribution. The recovered wind Levy increments have a narrow center, negative skewness, and large excess kurtosis. A NIG law is therefore retained for simulation. Its excess kurtosis is close to the solar one, but the skewness is weaker. The wind difficulty is therefore not only marginal tail thickness; it is also the combination of heavy-tailed innovations with a slow real factor.

Table 20: Fitted moments for recovered hourly Levy increments in the German wind CARMA model.

Distribution	Mean	Std. dev.	Skewness	Excess kurtosis
Gaussian	-2.7840×10^{-4}	3.1517×10^{-1}	0.000	0.000
NIG	-2.7839×10^{-4}	2.9423×10^{-1}	-0.370	15.002

Table 21: Distribution parameters for the recovered German wind driver.

Distribution	Parameter	Value
Gaussian	μ	-2.7840×10^{-4}
Gaussian	σ	3.1517×10^{-1}
NIG	α	1.5339
NIG	β	-8.5117×10^{-2}
NIG	δ	1.3218×10^{-1}
NIG	μ	7.0675×10^{-3}

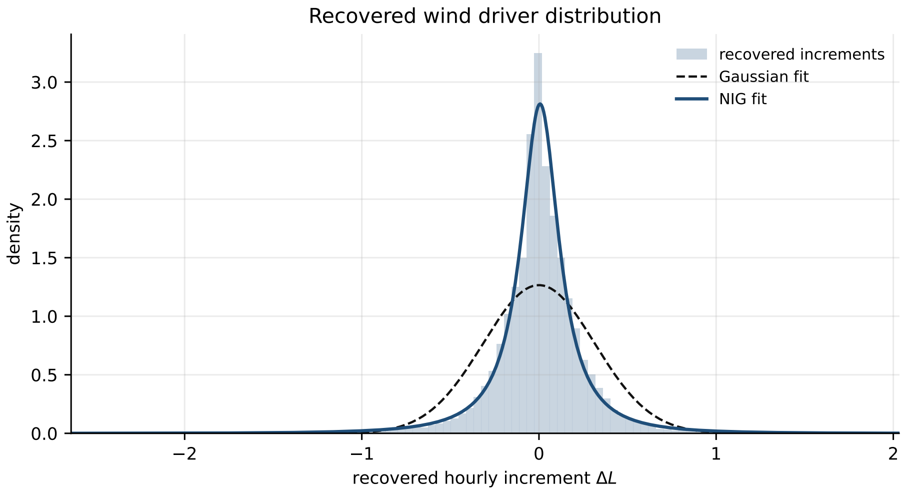

Figure 23: Recovered hourly Levy driver increments for the German wind CARMA model. The NIG fit captures the narrow center and heavy tails better than the Gaussian fit.

Figure 24 reports the induced distributions at the latent residual, residual-increment, and physical capacity-factor levels. The physical mapping adds the fitted logit seasonality and then applies the inverse logit transform.

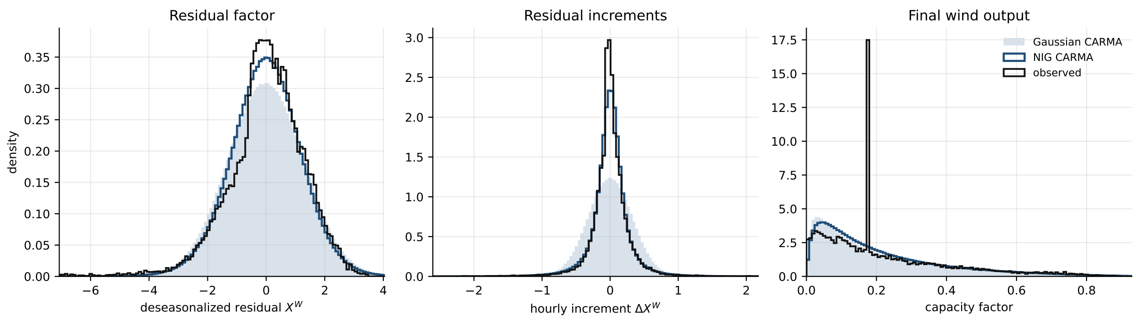

Figure 24: Distribution checks for the German wind marginal model at the deseasonalized logit residual, hourly residual-increment, and physical capacity-factor levels.

The NIG driver improves the residual-increment tails relative to a Gaussian driver, but the latent residual distribution remains more left-skewed in the data than in the simulated CARMA paths. This limitation is most visible in the logit coordinate and is partly compressed after mapping back to capacity factors.

4 Coupling Calibration

4.1 Temperature–Price Coupling

Figure 25 reports the joint cloud of the deseasonalized temperature residual and the deseasonalized German log-price residual. The dependence is not visually sharp, but the negative sample correlation is consistent with the winter heating channel: below-seasonal temperatures tend to increase electricity demand and spot prices. This relation is only a reduced-form diagnostic, since the coupled CARMA model acts on shocks rather than on residual levels.

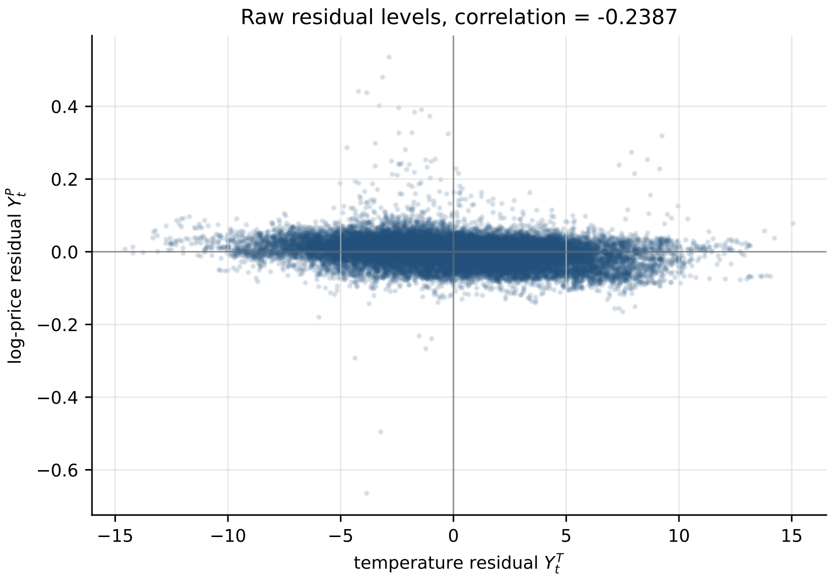

Figure 25: Raw residual relation between the deseasonalized German temperature residual and the deseasonalized German log-price residual.

Model. The marginal CARMA models from section 3 are kept fixed. Let Z^T be the temperature state and Z^P the price state. The one-way coupled model is

$$dZ_t^T = A_T Z_t^T\,dt + \sigma_T e_T\,dW_t^T, \qquad Y_t^T = b_T^\top Z_t^T, \tag{30}$$

$$dZ_t^P = A_P Z_t^P\,dt + \lambda\,\sigma_T e_P\,dW_t^T + e_P\,dL_t^{P,\mathrm{id}}, \qquad Y_t^P = b_P^\top Z_t^P. \tag{31}$$

The parameter λ multiplies the common temperature shock in the price driver. It does not modify the price drift or the deterministic seasonality. A positive shock in Y^T is a warmer-than-seasonal temperature shock. Hence a negative λ means that warmer-than-seasonal shocks lower price shocks, or equivalently that colder-than-seasonal shocks raise price shocks. This is the expected winter heating sign, since cold weather increases electric heating demand. The sign is not structurally constant over the year: in summer, warmer-than-seasonal shocks can increase cooling demand and therefore raise prices. The pooled estimate should therefore be read as an average shock transmission over the calibration window.

Estimator. For one-hour intervals, filtered state residuals are computed from the fixed marginal models:

$$\widehat{R}_i^T = \widehat{Z}_{t_i+\Delta}^T - \exp(A_T\Delta)\widehat{Z}_{t_i}^T, \qquad \widehat{R}_i^P = \widehat{Z}_{t_i+\Delta}^P - \exp(A_P\Delta)\widehat{Z}_{t_i}^P. \tag{32}$$

Under the independence assumption $dW^T \perp dL^{P,\mathrm{id}}$, their cross-covariance satisfies

$$\mathrm{Cov}(R_i^T, R_i^P) = \lambda\,\sigma_T^2 \int_0^\Delta \exp(A_T u) e_T e_P^\top \exp(A_P^\top u)\,du. \tag{33}$$

With

$$\widehat{D}_{TP} = \frac{1}{n}\sum_{i=1}^{n}(\widehat{R}_i^T - \overline{R}^T)(\widehat{R}_i^P - \overline{R}^P)^\top$$

and K_{TP} denoting the integral in eq. (33) for $\lambda = 1$, the scalar estimate is obtained by projecting through the observable CARMA directions:

$$\widehat{\lambda} = \frac{b_T^\top \widehat{D}_{TP} b_P}{b_T^\top K_{TP} b_P}. \tag{34}$$

This gives

$$\widehat{\lambda} = -1.3689 \times 10^{-3}$$

on 26303 common hourly intervals.

Table 22: Temperature–price coupling diagnostics.

Quantity	Value
Raw residual correlation	-0.2387
Projected state-residual correlation	-0.0405
$\widehat{\lambda}$ from state-residual covariance	-1.3689×10^{-3}
$\mathrm{Corr.}(dL^{P,\mathrm{marg}}, dL^T)$	-0.0416
$\mathrm{Corr.}(dL^{P,\mathrm{id}}, dL^T)$	-0.0009
Pearson p-value after removal	0.8790

After removing $\widehat{\lambda} dL^T$, the linear correlation between the temperature driver and the idiosyncratic price driver is close to zero, and the zero-correlation hypothesis is not rejected by the Pearson test. This supports the independence assumption at the covariance level, while not proving full distributional independence.

Idiosyncratic driver. The recovered marginal price driver contains the historical temperature component. The driver used for joint simulation is therefore decomposed as

$$dL_i^{P,\mathrm{marg}} = dL_i^{P,\mathrm{id}} + \widehat{\lambda}\, dL_i^T, \qquad dL_i^{P,\mathrm{id}} = dL_i^{P,\mathrm{marg}} - \widehat{\lambda}\, dL_i^T. \tag{35}$$

Both a Gaussian law and a NIG law are fitted to $dL^{P,\mathrm{id}}$. The NIG law is retained for simulation.

Table 23: Distribution fits for the idiosyncratic price driver after removing the temperature component.

Component	Mean	Std. dev.	Log-likelihood	Comment
Gaussian $dL^{P,\mathrm{id}}$	1.3444×10^{-6}	1.3224×10^{-2}	76456.33	benchmark
NIG $dL^{P,\mathrm{id}}$	1.0837×10^{-6}	1.2127×10^{-2}	83475.92	retained

The transmitted temperature component is much narrower than the idiosyncratic price driver: the standard deviation of $\widehat{\lambda} dL^T$ is 5.3811×10^{-4}, about 4.4% of the fitted NIG standard deviation. Thus the coupling is detected in covariance diagnostics, but most of the marginal price-driver dispersion remains idiosyncratic.

4.2 Solar–Price Coupling

Figure 26 reports the same diagnostic for the solar latent residual. The solar factor is the deseasonalized clear-sky shortfall variable from section 2.3, so a positive value corresponds to a larger latent solar shortfall, not to higher solar production. The raw residual correlation is positive but small, which is the expected sign in this transformed coordinate. As for temperature, this is only a level diagnostic; the coupled model is estimated on the CARMA shocks.

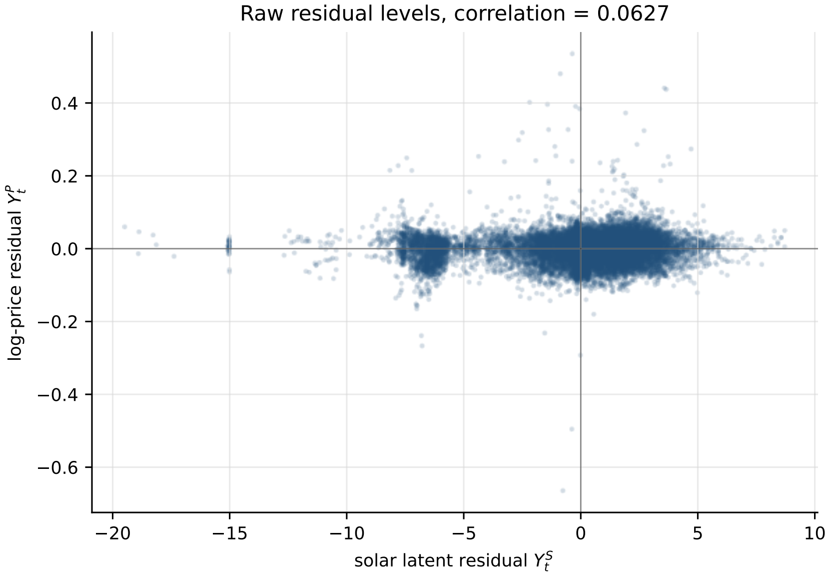

Figure 26: Raw residual relation between the deseasonalized German solar latent residual and the deseasonalized German log-price residual.

Model. The marginal solar and price CARMA models are kept fixed. Let Z^S be the solar latent state and Z^P the price state. The one-way coupled model is

$$dZ_t^S = A_S Z_t^S\,dt + \sigma_S e_S\,dW_t^S, \qquad Y_t^S = b_S^\top Z_t^S, \tag{36}$$

$$dZ_t^P = A_P Z_t^P\,dt + \lambda\,\sigma_S e_P\,dW_t^S + e_P\,dL_t^{P,\mathrm{id}}, \qquad Y_t^P = b_P^\top Z_t^P. \tag{37}$$

The parameter λ again acts only through the common shock. The sign must be interpreted in the transformed coordinate. Since Y^S is increasing with latent solar shortfall, a positive dW^S represents less effective solar availability, not more production. The merit-order argument therefore predicts a positive coupling coefficient: larger solar shortfall should increase residual demand and raise price shocks. Conversely, if the CARMA factor had been defined directly as above-seasonal production, the expected sign would be negative. The estimated coefficient below is slightly negative, but its magnitude is negligible and the driver correlations are statistically indistinguishable from zero. It is therefore not interpreted as an economic solar-price sign.

Estimator and diagnostics. The same covariance projection as in eq. (34) is applied with the solar CARMA matrices replacing the temperature matrices:

$$\widehat{\lambda} = \frac{b_S^\top \widehat{D}_{SP} b_P}{b_S^\top K_{SP} b_P}, \qquad \widehat{D}_{SP} = \frac{1}{n}\sum_{i=1}^{n}(\widehat{R}_i^S - \overline{R}^S)(\widehat{R}_i^P - \overline{R}^P)^\top. \tag{38}$$

This gives

$$\widehat{\lambda} = -7.9685 \times 10^{-6}$$

on 23103 common hourly intervals.

Table 24: Solar–price coupling diagnostics.

Quantity	Value
Raw residual correlation	0.0627
Projected state-residual correlation	-0.0018
$\widehat{\lambda}$ from state-residual covariance	-7.9685×10^{-6}
Corr.$(dL^{P,\mathrm{marg}}, dL^{S})$	-0.0010
Pearson p-value before removal	0.8790
Corr.$(dL^{P,\mathrm{id}}, dL^{S})$	0.0008
Pearson p-value after removal	0.9059

The zero-correlation hypothesis is not rejected even before removing the estimated solar component, and it remains clearly not rejected afterwards. Hence the solar coupling is negligible at the shock level in this calibration. This contrasts with temperature, where the raw and marginal-driver correlations are weak but still statistically detectable before the decomposition.

Idiosyncratic driver. The simulation identity is

$$dL_i^{P,\mathrm{marg}} = dL_i^{P,\mathrm{id}} + \widehat{\lambda}\, dL_i^{S}, \qquad dL_i^{P,\mathrm{id}} = dL_i^{P,\mathrm{marg}} - \widehat{\lambda}\, dL_i^{S}. \tag{39}$$

As for temperature, the NIG law is fitted to the idiosyncratic price driver rather than to the marginal price driver.

Table 25: Distribution fits for the idiosyncratic price driver after removing the solar component.

Component	Mean	Std. dev.	Log-likelihood	Comment
Gaussian $dL^{P,\mathrm{id}}$	2.5179×10^{-7}	1.2721×10^{-2}	68051.53	benchmark
NIG $dL^{P,\mathrm{id}}$	2.4479×10^{-7}	1.1613×10^{-2}	74016.55	retained

The transmitted solar component has standard deviation 2.2641×10^{-5}, about 0.2% of the fitted NIG standard deviation. It is therefore negligible for the marginal dispersion of the price driver. The solar marginal remains necessary for quanto payoffs through the payoff definition, but this calibration does not identify a material linear shock transmission from solar to spot prices.

4.3 Wind–Price Coupling

Figure 27 shows the raw relation between the wind logit residual and the German log-price residual. The wind transform is monotone in the capacity factor, so a positive wind residual corresponds to above-seasonal wind production. In contrast with solar, the level relation is visually and statistically clear: high wind residuals are associated with lower price residuals. This is consistent with the merit-order effect, since above-seasonal wind production reduces residual demand and therefore puts downward pressure on spot prices.

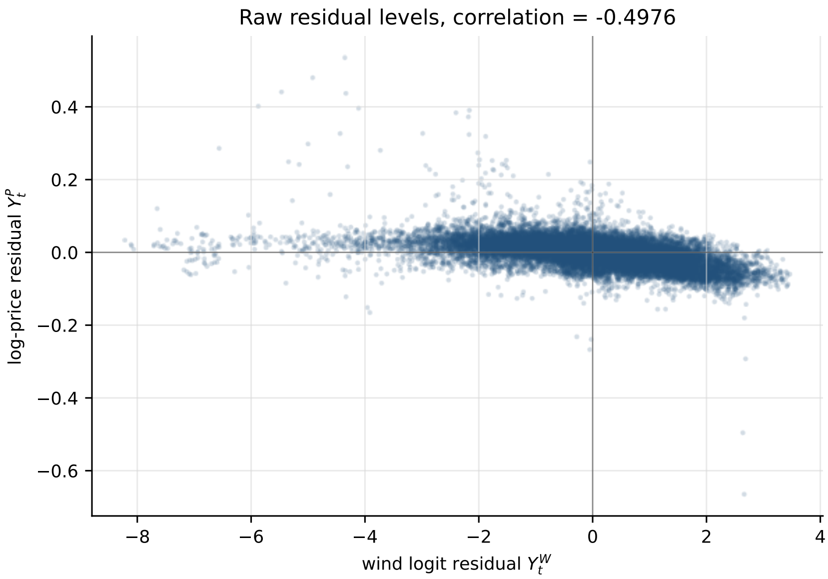

Figure 27: Raw residual relation between the deseasonalized German wind logit residual and the deseasonalized German log-price residual.

Model. The marginal wind and price CARMA models are kept fixed. Let Z^W be the wind state and Z^P the price state. The one-way coupled model is

$$dZ_t^W = A_W Z_t^W \, dt + \sigma_W e_W \, dW_t^W, \qquad Y_t^W = b_W^\top Z_t^W, \tag{40}$$

$$dZ_t^P = A_P Z_t^P \, dt + \lambda \, \sigma_W e_P \, dW_t^W + e_P \, dL_t^{P,\mathrm{id}}, \qquad Y_t^P = b_P^\top Z_t^P. \tag{41}$$

The parameter λ is a shock-transmission coefficient. It does not change the price drift. Because the wind logit transform is increasing in the physical capacity factor, a positive dW^W is an above-seasonal wind production shock. The expected merit-order sign is therefore negative: higher wind production lowers residual demand and is transmitted as a negative price shock. This interpretation is direct for wind, unlike solar where the latent coordinate is a shortfall variable.

Estimator and diagnostics. Using the same state-residual covariance projection,

$$\widehat{\lambda} = \frac{b_W^\top \widehat{D}_{WP} b_P}{b_W^\top K_{WP} b_P}, \qquad \widehat{D}_{WP} = \frac{1}{n} \sum_{i=1}^{n} (\widehat{R}_i^W - \overline{R}^W)(\widehat{R}_i^P - \overline{R}^P)^\top. \tag{42}$$

This gives

$$\widehat{\lambda} = -1.6913 \times 10^{-3}$$

on 26304 common hourly intervals.

Table 26: Wind–price coupling diagnostics.

Quantity	Value
Raw residual correlation	-0.4976
Projected state-residual correlation	-0.0402
$\widehat{\lambda}$ from state-residual covariance	-1.6913×10^{-3}
Corr.$(dL^{P,\mathrm{marg}}, dL^W)$	-0.0402
Pearson p-value before removal	6.6602×10^{-11}
Corr.$(dL^{P,\mathrm{id}}, dL^W)$	3.6858×10^{-5}
Pearson p-value after removal	0.9952

The wind channel is therefore identified more clearly than the solar channel. The marginal price driver has a small but statistically significant negative correlation with the wind driver. After removing $\widehat{\lambda} dL^W$, the remaining idiosyncratic price driver has no detectable linear correlation with the wind driver.

Idiosyncratic driver. The coupled simulation identity is

$$dL_i^{P,\mathrm{marg}} = dL_i^{P,\mathrm{id}} + \widehat{\lambda}\, dL_i^W, \qquad dL_i^{P,\mathrm{id}} = dL_i^{P,\mathrm{marg}} - \widehat{\lambda}\, dL_i^W. \tag{43}$$

The price driver law used in joint simulation is again fitted after removing the wind component.

Table 27: Distribution fits for the idiosyncratic price driver after removing the wind component.

Component	Mean	Std. dev.	Log-likelihood	Comment
Gaussian $dL^{P,\mathrm{id}}$	1.4198×10^{-6}	1.3225×10^{-2}	76458.21	benchmark
NIG $dL^{P,\mathrm{id}}$	7.3706×10^{-7}	1.2115×10^{-2}	83508.06	retained

The transmitted wind component has standard deviation 5.3305×10^{-4}, about 4.4% of the fitted NIG standard deviation. This is close to the relative size of the temperature component, while solar is two orders of magnitude smaller. Among the renewable factors, wind is therefore the only one for which the current calibration detects a material linear shock-transmission channel to German spot prices.

5 Pricing

6 Conclusion

References